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# Tell me a story! Narrative-driven XAI with Large Language Models

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ABSTRACT

Existing Explainable AI (XAI) approaches, such as the widely used SHAP values or counterfactual (CF) explanations, are arguably often too technical for users to understand and act upon. To enhance comprehension of explanations of AI decisions and the overall user experience, we introduce XAIstories, which leverage Large Language Models (LLMs) to provide narratives about how AI predictions are made: SHAPstories based on SHAP and CFstories on CF explanations. We study the impact of our approach on users' experience and understanding of AI predictions. Our results are striking: over 90% of the surveyed general audience finds the narratives generated by SHAPstories convincing, and over 78% for CFstories, in a tabular data experiment. More than 75% of the respondents in an image experiment find CFstories more or equally convincing as their own crafted stories. We also find that the generated stories help users to more accurately summarize and understand AI decisions than they do when only SHAP values are provided. The results indicate that combining LLM generated stories with current XAI methods is a promising and impactful research direction.

## 1. Introduction

A growing research area and practical challenge is to improve the transparency of AI models [1–3]. Explainable AI (XAI) aims at making the underlying decision mechanisms of 'black box' models understandable, proving to be useful for justifying model predictions towards data subjects, understanding misclassifications, improving a model's reliability in real-life applications, and increasing users' trust in the model [1]. However, state-of-the-art XAI methods do not generate concise, clear, natural language-based explanations, with a narrative that can possibly further help users.

For instance, consider the case of the popular SHAP (SHapley Additive exPlanations) method [4]. An example of a SHAP explanation for a loan application that is rejected by an AI model is given in Fig. 1. The features include various credit and loan applicant characteristics. The reader is invited to interpret the figure and create a narrative for the applicant about why the AI model predicts poor creditworthiness — we revisit this example in Section 4. This small exercise shows that current XAI explanations are difficult to understand, being too technical, unlike simple narratives, particularly for people with less technical training.

In this paper, we address the need to generate natural language explanations that provide a narrative that interprets an XAI explanation.[2] Our contributions are threefold:

1. We introduce and motivate a new XAI approach, termed the XAIstory, opening a new research direction for explainable AI.
2. We provide a first LLM-based method to generate narratives for two commonly used XAI methods: counterfactuals and SHAP values.
3. We show, using six studies, that the method leads to an improved user experience (in terms of convincingness, ease of use, confidence and likeliness to use) and comprehension of the explanations provided.

The paper is structured as follows. Section 2 positions this research in the current literature, which also motivates the need for XAIstories. Section 3 details the definition and methodology behind XAIstories. Next, Section 4 describes the output of XAIstories and the results derived from user study surveys, while Sections 5 and 6 conclude the paper with discussion and suggestions for future work.

## 2. Related literature

### 2.1. The need for explanations

An explanation is an attempt to convey the internal state or logic of an algorithm that leads to a decision [5]. Prior work suggests that when

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[2] The curious reader may already look at Fig. 4 for a generated XAIstory.

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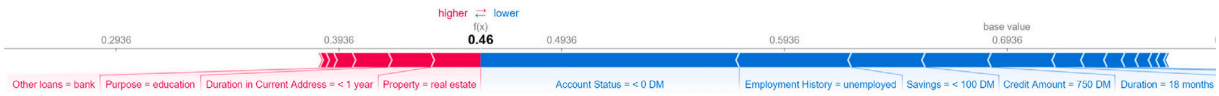


**Fig. 1.** Example of a SHAP explanation for a rejected (0) loan application.

users do not understand the workings of AI models and become more skeptical and reluctant to use the model, even if the model is known to improve decision performance [6–9]. The impact of explanations for example on trust, user acceptance, learning or decision making, has also been studied in the context of knowledge based systems [7,10–13].

The need for explainability is not new. Michalski introduced the 'comprehensibility postulate' in 1983 [14], arguing that "results of computer induction should be symbolic descriptions [...] directly interpretable in natural language". However, current XAI research has largely ignored this call, focusing instead on more structured, technical solutions which, as indicated by the exercise with Fig. 1, are challenging to understand.

The reason for this challenge can be found in what constitutes a 'good' explanation: one that also improves the *mental fit* between the AI model and the mental model of a person receiving the explanation [8,13,15], which also depends on factors such as the user's expertise [7,11]. For computer science experts, such technical solutions can improve their own mental fit, also often leading to improving the AI models themselves. Yet, not for non-technical persons, such as most business experts or decision subjects of AI models, where natural language is found to be the preferred manner to communicate scientific findings [16]. This mismatch between current technical XAI solutions and human decision-making, known as the 'inmates running the asylum' phenomenon [17], arguably occurs because AI researchers often create solutions tailored to their own needs rather than to those of non-expert users. A seminal joint study of academics and practitioners similarly concluded [18]: "There is thus a gap between explainability in practice and the goal of transparency, since explanations primarily serve internal stakeholders rather than external ones". So far, the gap remains.

### 2.2. Explanation methods

There are two sorts of explanation procedures for prediction models: global explanations, which provide an understanding for the model's overall behavior across all (training) instances [15,19], and instance-level explanations, which explain the model's prediction for a single instance [2,15,20]. As most often we need to explain single predictions, the focus of current XAI research is largely on instance-based explanations, with feature attribution and counterfactual (CF) methods being two popular solutions [2,19].

A CF explanation provides a minimum set of evidence present in the data instance to be explained (also called the factual), such that removing that evidence would change the decision [5,15]. In recent years, various techniques have been proposed to create such explanations, for tabular [5], image [21], and textual data [19,22]. LIME and SHAP are feature attribution methods that provide a set of feature-value combinations with weights that indicate their importance in the prediction [4,19], applicable to various data types.

### 2.3. Motivation for narrative-based explanations

The literature already points to the importance and impact of the structure of explanations [23] and the need for *narrative-based* solutions.

First, to overcome the mismatch between the technical solutions provided by methods like SHAP, and the mental model of domain experts and lay end users, narratives from XAI outputs are typically already created manually, with less ease and control in the latter case. For example, De Lange et al. (2022) provide narratives to describe SHAP waterfall plots in financial intermediation [24], while Zhou et al. (2023) enhance SHAP explanations with narrative descriptions in a fraud detection application [25]. Similarly, narratives are important in employment and real estate applications, as shown by De Oliveira et al. (2023) and Varun et al. (2020), helping to make AI models easier to adopt and more practical to use [26,27].

Second, the fields of communication theory and psychology also point to the benefits of a narrative-based approach to explanations. For example, narratives are naturally more accessible and memorable, making them particularly effective when communicating scientific evidence to a non-expert audience [16]. Narratives can also lower the cognitive load on users, making it easier for them to relate abstract model elements to real-world instances [28]. Research also suggests that narrative explanations are processed more quickly than logical-scientific descriptions by non-expert audiences, regardless of their prior knowledge of the topic [29]. Narratives were found to be more beneficial for users with lower familiarity with conceptual modeling [28]. Moreover, narratives tend to be more engaging and persuasive, enhancing trust and understanding of AI models among non-experts [16,30].

Third, there are also legal reasons for the use of narratives when explaining AI decisions. For instance, under European GDPR Article 12 and Recital 58, the principle of transparency requires "*concise, easily accessible and easy to understand, and that clear and plain language [...] be used*" [31,32].

Our work on XAIstories is also in line with the general research agenda in human-AI interactions [9], which emphasizes the need for designs that *"improve decision-making performance"* by effectively aligning AI capabilities with human decision making processes. By automatically developing narrative explanations that make AI decisions more comprehensible, our research directly contributes in this direction.

Finally, we note there is recent related work developing at the intersection of XAI and LLMs. For example, TalkToModel is an interactive dialogue system that chooses an XAI method and returns an explanation in textual form [33]. The scope of TalkToModel, and other such work, is however different: it interprets a question, maps it to appropriate explanations, and generates a textual response to the question. The last step does not create narratives about AI predictions, but rather simpler responses such as how important a feature is, what the most important features are, or what a counterfactual explanation may be. XAIstories, on the other hand, provide more complete narratives about AI predictions.

## 3. Methodology

### 3.1. Narratives generation

We combine XAI methods, specifically SHAP values or CF explanations, with LLMs. Given a machine learning model, an input, the output of the model for that input, and an XAI explanation for that model output, we construct different types of prompts that we provide to an LLM which then provides a narrative. We study how different prompt strategies affect the narratives, and the reactions of users to these narratives, using six surveys. As this is a first approach to generate XAI narratives with LLMs (as far as we know), the results are conservative: better prompting or LLMs will likely further improve the value of this approach.

**Prompt Types**: We present two types of prompts (Fig. 2), agnostic to the AI or LLMs used: *LLMstories*, that include the input instance and prediction to be explained, along with a description of the features; *XAIstories*, that additionally include an XAI explanation (SHAP or CF)

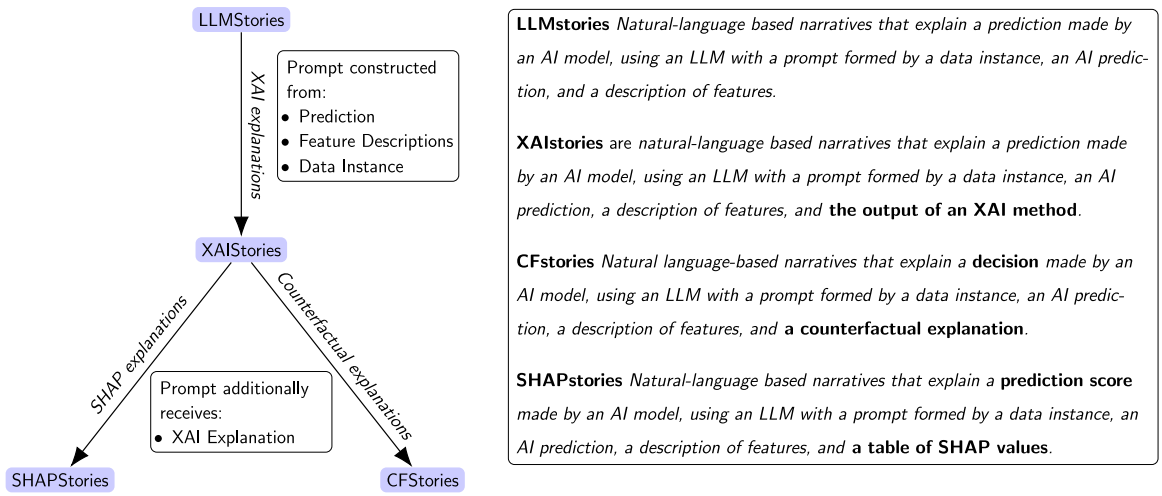


**Fig. 2.** Taxonomy diagram and definitions for LLM, XAI, SHAP and CF-stories.

- see below for more details. We then implement two special cases of XAIstories; SHAPstories and CFstories, which integrate SHAP and Counterfactuals explanations respectively in the prompt.

**LLM Used**: We use GPT-4 as it is currently widely recognized as a well-performing LLM able to process both textual and visual inputs [34].[3] Furthermore, it requires no fine-tuning for specific tasks, is accessible via an API, and has integrated plug-ins.

### 3.2. Data and implementation

**Tabular data:** The prompt to generate *SHAPstories* (*CFstories*) consists of several components: (a) outline of the target variable and all input features, detailing the meaning of each feature (commonly available); (b) an explanation of how SHAP (CF) works; (c) whether the model's classification was accurate; (d) a request for the LLM to come up with a plausible, fluent story as to why the model could have predicted the specific outcome. For *SHAPstories*, the LLM is explicitly required to only use the most important features. This addresses the limitation of SHAP, which assigns importance to all features [2]. As CF explanations only use a few features, there is no need to add this restriction for *CFstories*. The LLM is also asked to end with a summary and limit its response to eight sentences maximum, which should enhance the persuasiveness of narratives [16]. Finally, for *SHAPstories* the prompt has a table with the value and SHAP value of each feature of the instance, while for *CFstories* the CF explanation is provided. For all tabular experiments, we train a Random Forest classifier using cross-validation and RandomSearchCV using the Python scikit-learn package [35]. We use treeSHAP [4,36] for SHAP calculations, and DICE [37] for CF generation.[4]

**Image Data:** we study narratives for image data, using counterfactual explanations.[5] This allows us to also explore the approach for non-tabular data. We use Semantic Counterfactuals for Accurate Picture (SCAP) explanations [38], which identify semantically meaningful segments within an image that, when altered (in our case, blurred), results in a change in the predicted class. This approach allows us to integrate automatically described (labeled) segments into a prompt as: "The image was initially classified as [Initial Class], however, when [Counterfactual Segment] was blurred, the model changed its classification to [Counterfactual Class]".

We also use SceneXplain for GPT-4 [39], which represents an image as a detailed text description. By adding this description to the prompt, we are able to provide a LLM with information about the image without providing the image itself. In our experiments, a pre-trained Vision Transformer (ViT) [40], fine-tuned on Imagenet-1k, is the classifier we aim to explain. We then present narratives for images misclassified on the COCO dataset [41]. We focus on misclassified images only as the counterfactuals for the correctly classified ones are trivial for these images. For example an image of a dog being correctly predicted to be a dog, has a trivial counterfactual segment to be the dog in the picture.

**Data:** The following available (Kaggle and UCI) datasets are selected based on their features that allow semantic interpretation, their typical number of features of around 25 to 40, and because they originate from different domains:

- FIFA World Cup 2018 Man of The Match dataset [42], to predict whether a football team will have the 'Man of the Match' winner in a FIFA 2018 world cup match, based on the team's statistics.
- Student Performance dataset [43], to predict whether a student in secondary education will pass or fail mathematics.
- German Credit Scoring Dataset [44], to predict creditworthiness of loan applicants based on the loan and applicant characteristics.

### 3.3. Survey design

In our surveys, the participants are presented cases for which an AI model made a (stated) correct or incorrect prediction, and a description of the dataset and instance itself. We conducted a total of six surveys (Fig. 3), differing in three dimensions: (1) target audience: lay users vs data scientists, (2) criteria: user experience versus decision making and understanding, and (3) underlying explanation: SHAP vs CF.

**Target audience:** The lay users are questioned through Amazon Mechanical Turk (AMT), providing 4 US$ per user, requesting AMT 'Masters', and High School Graduate, each time asking for 70 users (in line with other studies, cf. [45]). An attention question was added in all surveys, users that answered that wrong were excluded from the results (explaining the final number of users – those with valid responses – $N$ not always being 70). Only for images, the task was less time consuming and more invalid responses were obtained, so a total of 160 users was

[3] We also experimented with Google's Gemini and Meta's Llama, but anecdotally found the generated stories less clear and convincing. We made our implementation with API calls and wrappers that can be reused with future LLMs, and can be found on GitHub [Link redacted for blinded review].

[4] All experiments are run on Google Colab. CFstories are generated in ChatGPT's playground, whilst SHAPstories use the API. The trained models, prompts, data and a full implementation for SHAPstories used in our experiments can be found at [redacted for blinded review].

[5] We did not consider SHAPstories for image data as for this type of data it is not obvious how to describe the typical heatmap type explanations in a natural language. This can be an interesting methodological future research direction.

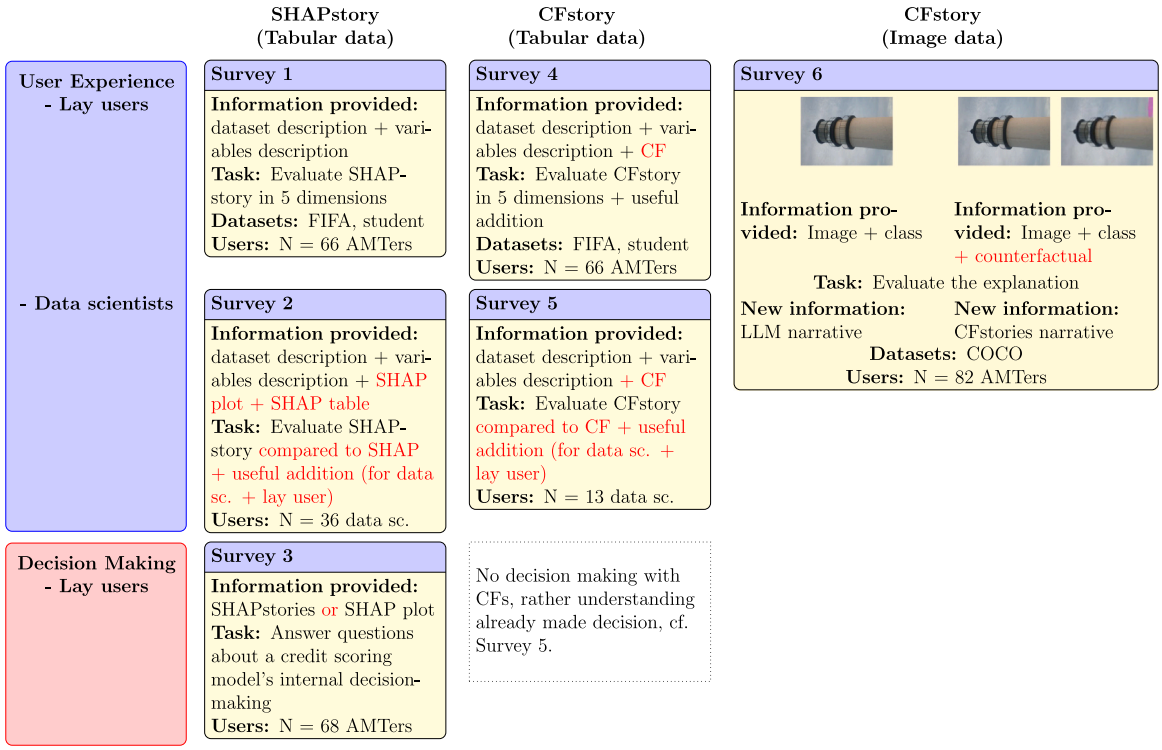


**Fig. 3.** Description of our survey designs for CFstories and SHAPstories.

asked, yielding $N = 82$ AMT users that provided valid responses. For the data scientists, the research community of the authors were queried (excluding those of our own research group), at two different times.

**Task evaluation:** The participants are asked to assess the provided story in 5 dimensions. The first is a Yes/No question, dimensions 2–5 ask participants to evaluate the statement on a 1–5 Likert scale (similar to [33]):

1. **Convincing**: Do you think this story provides a convincing interpretation of the model's prediction?
2. **Ease**: I felt the narratives were easy to interpret.
3. **Confidence**: I felt more confident in my understanding of the prediction with the narratives than without the narratives.
4. **Speed**: I felt I was able to arrive more rapidly at an understanding of the prediction with the narratives than without the narratives.
5. **Likeliness to use**: Based on my experience with the narratives, I consider them a valuable addition to AI models, that I would like to use.

When the data scientists are surveyed (Surveys 2 and 5), we additionally ask if the XAIstory provides a useful addition, compared to the XAI solution itself (be it SHAP in Survey 2 or CF in Survey 5).

In Survey 3, where we evaluate the decision making and comprehension improvement with SHAPstories, we quantitatively evaluate whether SHAPstories can improve the understanding of AI decisions. The study is based on a credit scoring decision making setting where ground truth for the comprehension questions asked is available (se Table 1), covering basic understanding as well as a hypothetical decision (question 4). Each of the questions is presented twice, once showing a SHAPstory and once showing a SHAP plot. Evaluation criteria are based on Huysmans et al. (2011) [46] who also assesses decision quality in an AI context. Since the ground truth is known for the first 3 questions, this setup allows us to evaluate the respondents' accuracy when answering the questions. We also record the time needed for the respondents to answer each question. A total of 8 questions are presented to the respondents (4 SHAP plots and 4 SHAPstories), where each question is about a different classification instance.

**Table 1**
SHAPstories' survey questions, describing which questions were asked to the respondents, both for SHAPstories and SHAP plots.

| Number | Question |
|---|---|
| 1 | Which of the following one-sentence summaries best describes why this individual gets a **poor/good** creditworthiness assessment? |
| 2 | Imagine feature **X** and feature **Y increase/decrease**. How is this likely to affect the model's decision? |
| 3 | Select the statement that corresponds to the **SHAP story/plot**. |
| 4 | Position yourself in the position of a banking employee. Your bank is only accepting clients with above-average, high, creditworthiness. Which credit decision would you make based on the above **SHAP story/plot**? |

**Explanation type:** Beyond a difference in what the SHAP and CF explanation looks like, an important distinction is that CFs are already close to a natural language. So for lay users, we provide the CF as well as the other input. For SHAP we do not do so, as users would not be able to understand this (we do so for data scientists, who should understand).

Finally, as a CF explains a decision while a SHAP explains a prediction score [47], the issue with CFs is less about a human *making* correct decisions with the explanation (as with SHAP), but more about *understanding* the explanation. This is already addressed in Survey 4 and motivates the lack of a survey on decision making with CFs.

In Survey 6, we switch to images, to study the applicability of the XAIstory approach beyond tabular data. AMTers are asked to come up with a two-line narrative, explaining what may have caused a provided misclassified image. Following this, we provide the respondents with an alternative narrative, generated by CFstories, and ask them to compare it with their own.

## 4. Results

We first provide some examples to illustrate different automatically generated narrative explanations, and then discuss the survey results.

**Prompt**

**Explanation goal:** Description of goal.

**SHAP description:** Brief explanation on the meaning of SHAP values.

**Dataset context:** Description of dataset, target, task and each feature.

**Feature and SHAP values:**

| Feature | Feature val | SHAP |
|---|---|---|
| Account Status | < 0 DM | -0.12 |
| Employment | Unemployed | -0.05 |
| Savings | < 100 DM | -0.03 |
| | ... | |

**Prediction description:** The model predicted that the individual is not credit worthy with a 46% confidence.

**SHAPstory (Narrative)**

*Our individual is a young, unemployed person with a significant savings amount less than 100 DM (Deutsche Mark). They reside in their own house and have lived at their current address for less than a year. Their current account status indicates a balance less than 0 DM which appears as the largest negative contributor to the prediction. [...]*
*In summary, while the individual did have some positive factors in their favor, their account status, employment status, savings amount, and recent credit amount tipped the balance towards more risk for the lender, causing the model to predict them as more likely 'not creditworthy'. The decision seems to underscore the AI's emphasis on an individual's financial stability and income sources when predicting creditworthiness.*

**Fig. 4.** Example SHAPstory for the credit scoring dataset, cf. the SHAP plot found in Fig. 1.

**Prompt**

**Explanation goal:** Description of goal.

**Counterfactual Changes:** If the student's number of absences decreased from 54 to 8, then the model's prediction would change from Fail to Pass.

**Feature Values:**

| Absences | Age | Study Time | ... |
|---|---|---|---|
| 54 | 16 | 2 to 5 hours | ... |

**Dataset context:** Description of dataset, target, task, and each feature.

**Prediction description:** The model predicted that the student will fail with a 64.00% confidence.

**CFstory (Narrative)**

*The AI model predicted that the student would fail this year of school based on their home and school life details. In order to change the predicted outcome, a single counterfactual change was identified: the student's number of absences would need to decrease significantly from 54 absences to just 8.*
*Absences play a large role in academic success and predictions because they influence both the direct instruction time a student has and the student's ability to keep up with the material. In our case, the student has missed a significant proportion of classroom learning with 54 absences. This likely led the AI to predict failure as this is often correlated with such high absentee rates. [...]*
*In summary, the classification of this student as likely to fail was likely predominantly influenced by the student's high number of absences from school. The relationship between academic success and attendance is one that the AI model placed heavy importance on in its predictions*

**Fig. 5.** Example CFstory for the Student dataset.

### 4.1. XAIstories examples

**SHAPstory on tabular data:** A SHAPstories example is shown in Fig. 4 (incl. the prompt setup), based on the SHAP values example noted in Fig. 1 in the Introduction. The example is for a loan applicant who is predicted to have bad creditworthiness. The SHAPstories example seems understandable, natural and sufficiently detailed, likely to be satisfactory for non-data science experts [48].

**CFstory on tabular data:** A CFstory example is shown in Fig. 5 for a counterfactual of the Student mathematics performance dataset, where a student is correctly predicted to fail the course.

**CFstory on image data:** Table 2 shows two example images used to evaluate CFstories, along with the corresponding narratives provided by the CFstories. Presented with the counterfactuals, CFstories provide arguably better explanations with better reasoning. Baseline LLMstories, where the LLM is not provided the counterfactual explanations, are often unsuccessful in explaining the misclassifications as well.

### 4.2. User surveys results for tabular data

**User experience results**

The evaluation in terms of the five user experience dimensions, as well as the 'Useful addition' question, is shown in Table 3. For the dimensions on a Likert scale (from 'Ease' to 'Likeliness to use'), the percentage of users that Agreed or Strongly Agreed is shown. Larger values are indicated with a darker green, and statistical significance results (compared to 50%) are shown with asterisks.

At first glance the results are consistent: lay users strongly favor both SHAPstories and CFstories, while data scientists mainly see the value of the narratives for general audience usage. In terms of SHAPstories versus CFstories, the results for the former are better, which make sense as the CFs already are closer to natural language and therefore already are a better 'mental fit' as compared to the list of SHAP values, which describe a complex game-theoretical concept. Next, we zoom in on some results.

**Table 2**
Example of LLMstories, which does not have access to the CF explanations (indicated in purple), and CFstories, which does.

| | Image without CF | Image with CF | LLMstories | CFstories |
|---|---|---|---|---|
| Original classification: missile CF: cloud | | | *The classifier might have mistaken the elongated shape and structure of the lighthouse against the sky as the silhouette of a missile.* | *The cloud's shape and position in relation to the lighthouse might resemble the trail of a missile launch, causing the classifier to misidentify the image.* |
| Original classification: unicycle CF: person | | | *The AI image classifier might have misclassified the image as a unicycle due to the skateboard's single wheel visibility and the skateboarder's body position, which could resemble a unicycle rider.* | *The presence of a person, especially in dynamic poses, might lead the classifier to associate them with objects like unicycles due to similarity in balance and motion patterns often seen in unicycle performances.* |

**Table 3**
SHAPstories and CFstories results in terms of five user experience dimensions, and as a useful addition.

| | SHAPstories: general audience | SHAPstories: data science audience own usage | SHAPstories: data science audience: general audience usage | CFstories: general audience | CFstories: data science audience own usage | CFstories: data science audience: general audience usage |
|---|---|---|---|---|---|---|
| **Convincingness** | 93.2% *** | 77.8% *** | | 82.9% *** | 69.2% ** | |
| **Ease** | 92.4% *** | 61.1% * | 91.7% *** | 92.4% *** | 53.9% | 84.6% *** |
| **Confidence** | 79.7% *** | 38.9% | 91.7% *** | 76.7% *** | 37.8% | 84.6% *** |
| **Speed** | 87.5% *** | 44.4% | 88.9% *** | 69.9% *** | 30.8% | 76.9% *** |
| **Likeliness to use** | 92.2% *** | 61.1% * | 83.3% *** | 82.9% *** | 53.9% | 92.3% *** |
| **Useful addition** | | 81.4% *** | | 90.0% *** | 76.9% *** | |

*Note*: For all results, we test if the obtained result is statistically significant: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

*Convincingness:* The general audience (survey 1) finds the narratives generated by SHAPstories more convincing interpretations of the model's predictions in 93.2% of the cases. Non-data science experts, however, criticize the provided narratives for their limited set of attributes. From their viewpoint, this constraint leads to an incomplete portrayal of the factors influencing the AI model's predictions. This is due to the construction of the SHAPstories' prompt (focus on the top attributes only), and aligns with its goal of representing the most important information only.

For the data science audience (survey 2), we find similar positive results. In 77.8% of these evaluations, the data science experts agreed that SHAPstories convincingly explain the model's prediction. Furthermore, in 81.4% of the cases, these experts agree that the narratives produced by SHAPstories serve as a valuable addition to the SHAP plots in explaining the AI model's predictions.

*Ease, confidence, speed and likeliness to use:* Table 3 also further examines the four dimensions along which the respondents are asked to assess SHAPstories on a Likert-scale: 92.42% of the non-data scientist respondents agree or strongly agree with the statement that SHAPstories are easy to interpret, and 92.2% finds SHAPstories a valuable addition to AI models, that they are likely to use. Also, 87.5% of the non-data scientist respondents indicate that SHAPstories increase the speed with which they understand the AI model's prediction. Overall, these results suggest that SHAPstories are well-received in terms of user-friendliness, confidence, and efficiency, which bodes well for adoption and continued usage.

The second and third columns of Table 3 examine the same dimensions, but only for the data science experts. In this context, data science experts are asked to evaluate SHAPstories both for their own use, and for a broader non-expert audience. When considering its utility for their own usage, a majority (61.1%) of data science experts agree on the ease of use that SHAPstories provide. The data science experts also evaluate SHAPstories very positively having in mind the usage for a general audience. These experts indicate in 91.7% of the cases that SHAPstories will contribute to a non-specialist's ease and confidence in understanding an AI model's predictions. This is also reflected in the high likelihood that data scientists will use SHAPstories for a non-data scientists audience (83.3%).

*CFstories results:* A very similar result emerges when evaluating CFstories (Surveys 4–5). First, the general audiences finds the CFstories convincing, easy and fast to interpret, are confident in them and likely to use. Most interestingly, 90% of them find them a useful addition to the CF itself. All these numbers are statistically significant, but are also consistently lower than for SHAPstories.

The data science audience once more finds limited value for their own usage (even less than for SHAPstories), but also once more sees value for use by a general audience. Again with all statistically significant results, but overall lower than for SHAPstories, also confirming that CF explanations are more comprehensible than SHAP ones. These results indicate that the mental fit of lay users aligns closest with narratives, then counterfactual explanations, followed by SHAP explanations, confirming previous findings from communication theory [16].

*Qualitative feedback:* Finally, we gathered qualitative open ended feedback from the data science experts. We provide representative quotes indicating the potential of SHAPstories and CFstories.

*"They [SHAPstories] could be exceedingly helpful when it comes to evaluating model outputs by business users so that they can trust the model's output more easily. It could also help when formulating a rejection to a specific customer".*

18 respondents emphasize the utility of SHAPstories in scenarios where explanations of individual predictions are required for non-data science audiences. Examples include popular science communication, business communication, and storytelling for medical experts. Furthermore, they also highlight its effectiveness when multiple narratives are necessary.

*"It [SHAPstory] is easier to evaluate (and/or correct/modify) the correctness of a story against the numbers and the plots than it is to go through the numbers and construct the story from scratch".*

This quote was phrased only by one respondent, and indicates that SHAPstories can address some limitations of SHAP, namely that SHAP attributes significance to all features [2] and does not have an established protocol for interpretation [49].

*"It [SHAPstory] goes beyond explanation"* or *"[the narrative is] postulating beyond the facts".*

Five respondents emphasize that SHAPstories makes underlying assumptions that are not necessarily correct, a limitation we discuss below.

*"The added value is rather low as it mostly repeats the information from the plot".*

Six other respondents do not immediately see the added value of SHAPstories, stating that SHAP plot and tables already convey the same information. These respondents also mentioned that the SHAP plot and table provide more intuitive and efficient explanations. Earlier results, however, indicate that most of the data science experts refrain from such a critique when they have a non-data science audience in mind.

For CFstories the main two comments are the addition of information, but also the lack of information on other counterfactuals that exist.

*"It [CFstory] is too verbose and adds unnecessary details".*

*"It [CFstory] is one counterfactual that would have changes the decision of the model. But it does not provide insights into the other variables, which are also important to this decision".*

**Decision making and understanding**

Survey 3 studies to what extent the users' understanding of explanations are accurate. Accuracy is based on whether the respondents' answers to the questions in Table 1 agree with the correct answers to the questions — which are objective, as they are based on the machine learning model's behavior, and unique for questions 1–3. Regarding decision question 4 in Table 1, we measure whether the respondent's decision, while subjective, is the same as that of the machine learning model. A priori there is no reason why that would be the case. We also measure how long it takes for them to make a decision. The context is credit scoring.

*Accuracy:* Table 4 shows the percentage of correctly answered questions. Accuracy levels differ significantly depending on the question asked. For all first three questions, which ask respondents about their understanding of the internal-decision making of the model, SHAPstories scores the best. The differences are statistically significant ($p$-value $< 0.05$) for the first two questions. The difference relative to SHAP plots is the largest (50%) for question 1. SHAP plots, on the other hand, perform rather poorly indicating that the respondents do not understand well how the model has made a decision. The fourth question has the lowest score for both SHAPstories and SHAP plots, indicating that although people make different decisions than the model, they still understand the rationale of the model's decision as indicated by the first three questions.

**Table 4**
SHAPstories' survey results, describing the proportion of respondents correctly answering questions. Results reported separately for SHAPstories and SHAP plots, with p-values from a one-sided test on the population proportion. Significant results are indicated in **bold**.

| | Obs | SHAPstories | SHAP plots | Difference | p-value |
|---|---|---|---|---|---|
| Question 1 | 68 | **83.8%** | 33.8% | 50% | **0.000** |
| Question 2 | 68 | **64.7%** | 48.5% | 16.2% | **0.029** |
| Question 3 | 68 | 67.6% | 61.7% | 5.9% | 0.237 |
| *Question 4* | *68* | *44.1%* | *44.1%* | *0%* | *0.500* |

In line with all earlier results, these results further support that SHAPstories more accurately summarize and help users understand AI decisions.

*Speed:* The time (measured in seconds) needed for respondents to answer questions are on average consistently around 1 min, and do not significantly favor the stories or plots. On average, respondents are 10.6% slower when responding to questions based on SHAPstories. This increase in response time, however, improves the decision quality. To study this relationship further, we run two Poissons regressions, where accuracy and the confidence in their own answer serve as dependent variables. These are regressed on the time respondents need to answer a question. Our regression analysis shows a statistically significant positive relationship between the time taken to answer questions and accuracy for SHAPstories, suggesting that the additional time spent reading narratives significantly improves decision quality. Each second spent on narratives increased accuracy (coefficient $= 0.00059, p < 0.05$) without significantly affecting respondents' confidence in their answers (coefficient $= -0.00014, p > 0.1$). In contrast, for SHAP plots, each additional second did not significantly increase accuracy (coefficient $= 0.00061, p > 0.1$) but significantly reduced the respondents' confidence in their answers (coefficient $= -0.00045, 0.5 < p < 0.1$), indicating that more time on plots might confuse respondents rather than help them. These findings highlight the distinct impacts of narrative versus plot-based explanations on decision-making processes.

### 4.3. User survey results for image data

We compare CFstories and LLMstories along the dimensions of convincingness and speed, and relative to a baseline in which end users construct narratives themselves. We also measure how often CFstories and users' narratives include the counterfactuals.

**Convincingness:** 75.4% of all respondents find CFstories as convincing, or more convincing, than their own narrative. For the baseline LLMstories, this number drops to 66.8%. Both are statistically significant (relative to 50%) at 1% level for a one-tailed test on population proportion.

**Speed:** We also measure the time it takes for respondents to provide their responses. This allows us to compare the time to think of, and write a narrative, with the time they needed to simply read and evaluate the automatically generated narratives. On average, across all valid responses within the sample, respondents need 68 s to compose a narrative. In contrast, the average response time for evaluating the generated narratives is only 5 s, showing a significant speed advantage.

We also measure how often CFstories and users' narratives include the counterfactual explanations. While one would expect CFstories would always do so, there is no reason to guarantee the LLM would achieve this. In our data, CFstories narratives always included the given counterfactual, while only 79.7% of the narratives provided by the respondents contained the counterfactual (or a synonym).

In Table 5, we analyze the CFstories results for each survey question individually. For all images, except image 3, more than half of the respondents find CFstories' narratives at least as convincing as their own narratives. For question two, individuals exhibit the highest level of trust in CFstories' narrative. Here, the CFstory states: *"The presence of a person, especially in dynamic poses, might lead the classifier to associate*

**Table 5**
CFstories' survey results, describing which narrative users found more convincing, and a summary of keywords that users included in their narratives.

| | Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Question 6 |
|---|---|---|---|---|---|---|
| **Image** | | | | | | |
| | Class: missile<br>CF: cloud | Class: unicycle<br>CF: person | Class: rickshaw<br>CF: person | Class: goose<br>CF: sky | Class: blow dryer<br>CF: person | Class: soccer ball<br>CF: person |
| **Results** | (%)<br>Own 33.3<br>Similar 33.3<br>LLM 33.3<br>$p$-value: 0.0228 | (%)<br>Own 9.8<br>Similar 29.3<br>LLM 61.0<br>$p$-value: 0.0000 | (%)<br>Own 42.9<br>Similar 26.2<br>LLM 31.0<br>$p$-value: 0.1773 | (%)<br>Own 32.4<br>Similar 29.7<br>LLM 37.8<br>$p$-value: 0.0163 | (%)<br>Own 17.5<br>Similar 57.5<br>LLM 25.0<br>$p$-value: 0.0000 | (%)<br>Own 12.5<br>Similar 42.5<br>LLM 45.0<br>$p$-value: 0.0000 |
| **Keywords: CF provided** | clouds, (horizontal) orientation, shape, smoke/gasses | wheels, person (position, balance) | wheeled vehicle, two people (passenger, second rider, behind driver), umbrella, shadow | formation flying (symmetrical, leader), sky (blue, water resemblance, distant) | person (proximity to hair/face), cord/wire | person (goalkeeper, kneeling, dribbling) |
| **Keywords: CF not provided** | (horizontal/ sideways) orientation, shape (tall, thin, pointed, cylindrical), color, antenna | person (posture, balance, position), surface (curve), background (circle) | umbrella, rickshaw (wheels), two people (passenger, driver) | formation flying (flock), plane (size, color) | object (shape, cord/wire, handheld), person (proximity to hair/face) | person (stance/position, goalkeeper, kick), ball (flying), background (field, grass) |

Note: The displayed $p$-value are results for the tests on population proportion, grouping together individuals who indicated CFstories' narrative to be of similar quality or better than their own narrative.

*them with objects like unicycles due to the similarity in balance and motion patterns often seen in unicycle performances"*, incorporating keywords that also emerged in respondents' explanations, such as balance and position.

Finally, Table 5 gives insights into why the respondents may favor their own narrative, when that happens. In the cases where the CF is already provided, individuals who prefer their own story often tend to include additional characteristics that could have led to the misclassification in their narrative. For instance, in the 'missile' example, respondents added the orientation of the lighthouse as an additional reason for misclassification in their storytelling. One respondent described this as follows: *"The image was misclassified because of the horizontal orientation of the lighthouse making it appear it is heading through clouds like a missile would"*. These kinds of explanations also arise for the other images.

## 5. Discussion

### 5.1. Theoretical implications

Our findings support theories in cognitive and communication sciences that emphasize the effectiveness of narratives in reducing cognitive load and helping with processing of information [28,29] while increasing trust, and being more persuasive [16,30]. Indeed, our results indicate that XAIstories significantly improve user experience and understanding of AI models, particularly for lay users. By creating narratives for AI explanations, this research not only reinforces existing theories but also provides empirical evidence of their application in the field of explainable AI.

Our study also highlights the mental model differences between data scientists and lay users in interpreting AI explanations, in line with work on differences between how novices and experts process explanations [7,11]. More specifically, the results show lay users rate the narrative type explanations higher than data scientists do. Moreover, while data scientists (experts) find traditional SHAP plots valuable, they still recognize the utility of SHAPstories in communicating complex AI model outputs to a general audience. This reinforces the idea that explanations must be tailored to their intended users, as the mental fit between narrative-based and technical explanations varies.

Lastly, we introduced XAIstories as a novel approach in the XAI community. That in itself opens up a new research direction within the domain of XAI, as we discuss below, with potential new theoretical and methodological questions.

### 5.2. Practical implications

Our findings have several practical implications for organizations seeking to enhance the impact of their AI systems using XAI. First, companies can adopt narrative explanations like XAIstories to significantly improve user comprehension of AI models. The empirical evidence presented in this study demonstrates that narratives generated for SHAP and counterfactual explanations help users better understand the reasoning behind AI predictions compared to the more technical explanations, such as SHAP values or counterfactuals alone.

Second, XAIstories provide a concrete solution to the 'inmates running the asylum' problem by making existing AI explanations more accessible to non-expert users. By generating natural language-based explanations, XAIstories help bridge the gap between experts developing complex AI models and other stakeholders such as end users, facilitating a more human-centered approach to AI adoption and decision-making.

Finally, benefits of XAIstories do not come with risks of loss of compliance with related regulations. XAIstories align well with regulatory requirements for transparency, such as the European GDPR, which mandates that AI-based decision-making processes be explained in a clear and plain language. By automatically generating narrative

explanations that are both concise and comprehensible, XAIstories can help companies comply with these regulations and promote trust and ethical use of AI technologies.

### 5.3. Limitations and future work

While our study indicates the benefits of using automatically generated narrative explanations via LLMs, it has several limitations. One is the qualitative nature of the evaluation regarding behavioral impact. Our research primarily focuses on subjective measures, such as convincingness and ease of understanding. However, further studies are needed to explore how these narratives impact specific behaviors, including in-domain decision-making, risk-taking and adoption of AI. Understanding these changes across different contexts and objectives would provide further guidance on when and how to use narrative-driven XAI.

Another limitation involves the need for a more systematic, quantitative assessment of XAIstories. Future work should aim to develop methods to measure their accuracy, persuasiveness, and overall quality. Such metrics can also help in assessing the appropriate design choices in coming towards XAIstories, incl. what LLM to use. Studies based on field experiments can additionally shed light on the practical impact of the approach as well as the relation between the proposed metrics and actual behaviors and outcomes. Such metrics can also help automatically improve the stories themselves and potentially customize them to individual users.

There is also room to explore alternative approaches to generating XAIstories. While current implementations use prompt-based techniques with LLMs, future research could investigate agentic models, fine-tuning, or other natural language processing methods to create more tailored and context-sensitive narratives.

Finally, we foresee that XAIstories can be a starting point within an interactive dialogue-based setting, where users can further challenge the narrative or ask for additional evidence, once more opening up a new research direction within the Human-Computer Interaction domain, and in line with for example the TalkToModel appraoch [33].

## 6. Conclusion

Our study introduced XAIstories as a novel approach for enhancing the comprehensibility of AI models through narrative-driven explanations. The results are striking: in the image classification study, CFstories were as convincing as user-generated narratives in more than 75% of cases, while offering a tenfold time gain in understanding. For tabular data, over 90% of non-data scientists and 75% of data scientists found SHAPstories to be a convincing interpretation of AI model predictions, while for CFstories 90% of lay users, and 76% of data scientists, find it a useful addition over a CF explanation.

Previous literature has pointed out the limitations of current XAI solutions, often characterized as the 'inmates running the asylum,' where explanations are created by technical experts for their own use. XAIstories represent a first step in addressing these issues, offering a new solution that suggests the need to rethink what kinds of explanations we actually need. The impact of XAIstories could be substantial, acting as the missing link in XAI for effectively communicating model predictions to users. This includes not only the subjects impacted by model decisions but also domain experts who rely on AI models. At the same time, we foresee future research directions in both the qualitative and quantitative evaluations of XAIstories, exploring alternative methods for generating XAIstories, and extending their use to a range of contexts and applications.

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## CRediT authorship contribution statement

**David Martens:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **James Hinns:** Writing – review & editing, Software, Resources, Methodology, Conceptualization. **Camille Dams:** Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Data curation. **Mark Vergouwen:** Writing – review & editing, Visualization, Validation, Methodology. **Theodoros Evgeniou:** Writing – review & editing, Methodology.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT-4 to run the analyses, as detailed in the text, and chatGPT as a writing assistant to rephrase sentences. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

All data included is publicly available.

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